ELECTROMAGNETIC WAVES (EMW)

MATLAB Visualization

Visualization and Analysis of Vector Fields and Plane Electromagnetic Waves

Divergence, Curl and Gradient of the Electric and Magnetic Fields: A MATLAB Study

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Abstract

This report presents a computational study of the vector-field operators that underlie electromagnetic theory and of their behaviour for a uniform plane wave. Using MATLAB, three fields are analysed and visualised: (i) a purely rotational field $\mathbf{F} = (-y, x, 0)$, whose divergence is zero and whose curl is constant; (ii) a saddle-type field $\mathbf{F} = (x, -y, 0)$, which has zero divergence and zero curl; and (iii) a uniform plane electromagnetic wave travelling in an arbitrary direction $\mathbf{n}$ in free space.

For every field, the three-dimensional picture is complemented by two-dimensional views on the three coordinate planes. Divergence, curl and the gradient of the field magnitude are derived analytically, checked numerically, and verified through Stokes' theorem and the divergence theorem. For the plane wave, the electric field $\mathbf{E}$ and the magnetic field $\mathbf{H}$ are examined one by one, together with their curl, their divergence and the gradient of their magnitudes, and the wave is finally drawn in three dimensions together with the planes transverse to the direction of propagation.

The results confirm that $\nabla\cdot\mathbf{E} = \nabla\cdot\mathbf{H} = 0$, that $\nabla\times\mathbf{E}$ and $\nabla\times\mathbf{H}$ are in phase quadrature with the fields, in agreement with Faraday's and Ampère's laws, and that $\mathbf{E}$, $\mathbf{H}$ and $\mathbf{n}$ form a right-handed orthogonal triad.

Keywords: *divergence, curl, gradient, Stokes' theorem, Maxwell's equations, plane wave, Poynting vector, MATLAB.*

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1. Introduction

Divergence, curl and gradient are the three differential operators on which the whole of electromagnetic theory is built. Maxwell's equations are written entirely in terms of them, and the main properties of any field (its sources, its circulation and its ability to propagate as a wave) can be read directly from their values. Watching these operators act on concrete fields is therefore one of the most effective ways of building intuition before studying wave propagation, which is the central subject of the course.

1.1 Objectives

- Review the definition and the physical meaning of the divergence, the curl and the gradient.
- Visualise a vector field in three dimensions and on three orthogonal planes, and relate what is seen on each plane to the divergence and to the corresponding component of the curl.
- Analyse two test fields: a rotational field (non-zero curl) and a saddle field (zero divergence and zero curl), including checks with Stokes' theorem and the divergence theorem.
- Study a uniform plane electromagnetic wave travelling in a general direction **n**: plot **E**, $\nabla\times\mathbf{E}$, $\nabla\cdot\mathbf{E}$ and $\nabla|\mathbf{E}|$, and the same four quantities for **H**, each in 3D and on the three coordinate planes.
- Display the wave itself in three dimensions, including the planes perpendicular to the direction of propagation.

1.2 Organisation of the report

Section 2 summarises the mathematical background and the numerical and graphical method. Section 3 analyses the rotational field and Section 4 the saddle field. Section 5 applies the same tools to the uniform plane wave and presents the ten wave figures. Section 6 gives the conclusions.

1.3 Tools

All computations and figures were produced with MATLAB R2024a. The fields are sampled on regular grids created with *meshgrid*, the operators are evaluated with the built-in functions *divergence* and *curl*, and the figures are exported on a white background at high resolution (400–600 dpi).

2. Theoretical Background

2.1 Scalar fields, vector fields and the del operator

A **scalar field** $\psi(x, y, z)$ assigns a number to every point of space (a potential, or the magnitude of another field, for example), whereas a **vector field** $\mathbf{F}(x, y, z) = F_x\mathbf{a}_x + F_y\mathbf{a}_y + F_z\mathbf{a}_z$ assigns a vector, that is, a magnitude and a direction. The three operators of interest are conveniently written with the vector differential operator del (nabla):

$$\nabla = \mathbf{a}_x\,\partial/\partial x + \mathbf{a}_y\,\partial/\partial y + \mathbf{a}_z\,\partial/\partial z \tag{1}$$

2.2 Divergence

The divergence of a vector field is a scalar. It measures the net outward flux of the field per unit volume at a point:

$$\nabla\cdot\mathbf{F} = \lim_{\Delta v\to 0}\,(1/\Delta v)\oint_S \mathbf{F}\cdot d\mathbf{S} \tag{2}$$

In Cartesian coordinates it is obtained by adding the partial derivative of each component with respect to its own coordinate:

$$\nabla\cdot\mathbf{F} = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z \tag{3}$$

The divergence theorem connects the two descriptions: the volume integral of the divergence equals the flux through the closed surface that bounds the volume.

$$\int_V (\nabla\cdot\mathbf{F})\,dv = \oint_S \mathbf{F}\cdot d\mathbf{S} \tag{4}$$

A positive divergence indicates a **source** (field lines start there), a negative divergence indicates a **sink** (field lines end there), and a zero divergence indicates a **solenoidal** field, whose lines are continuous. In electromagnetics, $\nabla\cdot\mathbf{D} = \rho_v$ says that free charge is the source of electric flux density, while $\nabla\cdot\mathbf{B} = 0$ says that magnetic field lines never begin or end.

2.3 Curl

The curl of a vector field is another vector field. Its component along a unit vector $\mathbf{a}_n$ is the circulation of the field per unit area around a small loop whose normal is $\mathbf{a}_n$:

$$(\nabla\times\mathbf{F})\cdot\mathbf{a}_n = \lim_{\Delta s\to 0}\,(1/\Delta s)\oint_C \mathbf{F}\cdot d\mathbf{l} \tag{5}$$

In Cartesian coordinates the three components are:

$$\begin{aligned}(\nabla\times\mathbf{F})_x &= \partial F_z/\partial y - \partial F_y/\partial z\\ (\nabla\times\mathbf{F})_y &= \partial F_x/\partial z - \partial F_z/\partial x\\ (\nabla\times\mathbf{F})_z &= \partial F_y/\partial x - \partial F_x/\partial y\end{aligned} \tag{6}$$

Stokes' theorem relates the surface integral of the curl to the circulation along the contour that bounds the surface:

$$\int_S (\nabla\times\mathbf{F})\cdot d\mathbf{S} = \oint_C \mathbf{F}\cdot d\mathbf{l} \tag{7}$$

The direction of the curl follows the right-hand rule, and its magnitude is twice the angular velocity of a small paddle wheel placed in the field. A field with $\nabla\times\mathbf{F} = 0$ is **irrotational**: its line integral is independent of the path and it can be written as the gradient of a scalar potential, $\mathbf{F} = \nabla\varphi$.

> **Remark.** The component of the curl along an axis measures the circulation *in the plane perpendicular to that axis*. This is why, for the 2D plots, $(\nabla\times\mathbf{F})_z$ is the quantity that matters on the XY plane, $(\nabla\times\mathbf{F})_y$ on the XZ plane and $(\nabla\times\mathbf{F})_x$ on the YZ plane.

2.4 Gradient

The gradient of a scalar field is a vector that points in the direction of the steepest increase of the field, and its magnitude is the rate of increase in that direction. It is perpendicular to the surfaces ψ = constant:

$$\nabla\psi = \mathbf{a}_x\,\partial\psi/\partial x + \mathbf{a}_y\,\partial\psi/\partial y + \mathbf{a}_z\,\partial\psi/\partial z \tag{8}$$

In this report the gradient is applied to the magnitude of the wave fields, $\psi = |\mathbf{E}|$ and $\psi = |\mathbf{H}|$. Two identities are used repeatedly:

$$\nabla\times(\nabla\psi) = 0 \tag{9}$$
$$\nabla\cdot(\nabla\times\mathbf{F}) = 0$$

The first identity shows that every gradient field is irrotational; the second shows that every curl field is solenoidal. The Laplacian $\nabla^2\psi = \nabla\cdot(\nabla\psi)$ is the divergence of the gradient.

Table 1. Summary of the three operators.

Operator	Acts on	Result	Physical meaning
Gradient $\nabla\psi$	scalar	vector	Direction and rate of steepest increase
Divergence $\nabla\cdot\mathbf{F}$	vector	scalar	Net outward flux per unit volume (sources and sinks)
Curl $\nabla\times\mathbf{F}$	vector	vector	Circulation per unit area (rotation)
Laplacian $\nabla^2\psi$	scalar	scalar	Divergence of the gradient

2.5 Numerical evaluation

Each field is evaluated on a uniform grid generated with *meshgrid*, and the divergence and the curl are computed with MATLAB's *divergence* and *curl* functions, which use central differences at interior nodes and one-sided differences at the boundary. For the two test fields, whose components are linear in x and y, the finite-difference result is exact up to rounding. For the plane wave, whose components are sinusoidal, the numerical result agrees with the analytical expression up to a truncation error that decreases with the grid spacing; the script prints the maximum relative error, evaluated on interior nodes only so that boundary effects are excluded. Line integrals used for the Stokes and divergence-theorem checks are evaluated with the trapezoidal rule (*trapz*).

2.6 Graphical method

Every field is presented in four views: a three-dimensional view with the arrows and three semi-transparent coordinate planes, followed by three two-dimensional views, one on each plane (XY, XZ and YZ). Two points must be kept in mind when reading the 2D plots. First, an arrow plot on a plane shows only the two in-plane components; the third component, normal to the plane, cannot be drawn and is stated in the annotation. Second, as explained in the remark of Section 2.3, only the curl component normal to a plane describes circulation within that plane.

For the two test fields the planes are placed at $z = 0$, $y = 1$ and $x = 1$. The offset is deliberate: at $y = 0$ the in-plane vector $(-y, 0)$ would vanish and the XZ plot would be empty. For the plane wave the planes pass through the origin, and colour maps are used to show magnitudes and scalar quantities.

3. Case Study 1: A Rotational Field

3.1 Description of the field

The first field is

$$\mathbf{F} = -y\,\mathbf{a}_x + x\,\mathbf{a}_y + 0\,\mathbf{a}_z \tag{10}$$

Its magnitude is $|\mathbf{F}| = \sqrt{(x^2 + y^2)} = r$, the distance from the z-axis, and at every point the vector is tangent to the circle of radius r around the z-axis, pointing counter-clockwise. In cylindrical coordinates the field is simply $\mathbf{F} = r\,\mathbf{a}_\varphi$. It is the velocity field of a rigid body rotating about the z-axis with angular velocity $\boldsymbol{\omega} = \mathbf{a}_z$ rad/s, since $\mathbf{F} = \boldsymbol{\omega} \times \mathbf{r}$. The field is shown in three dimensions in Figure 1.

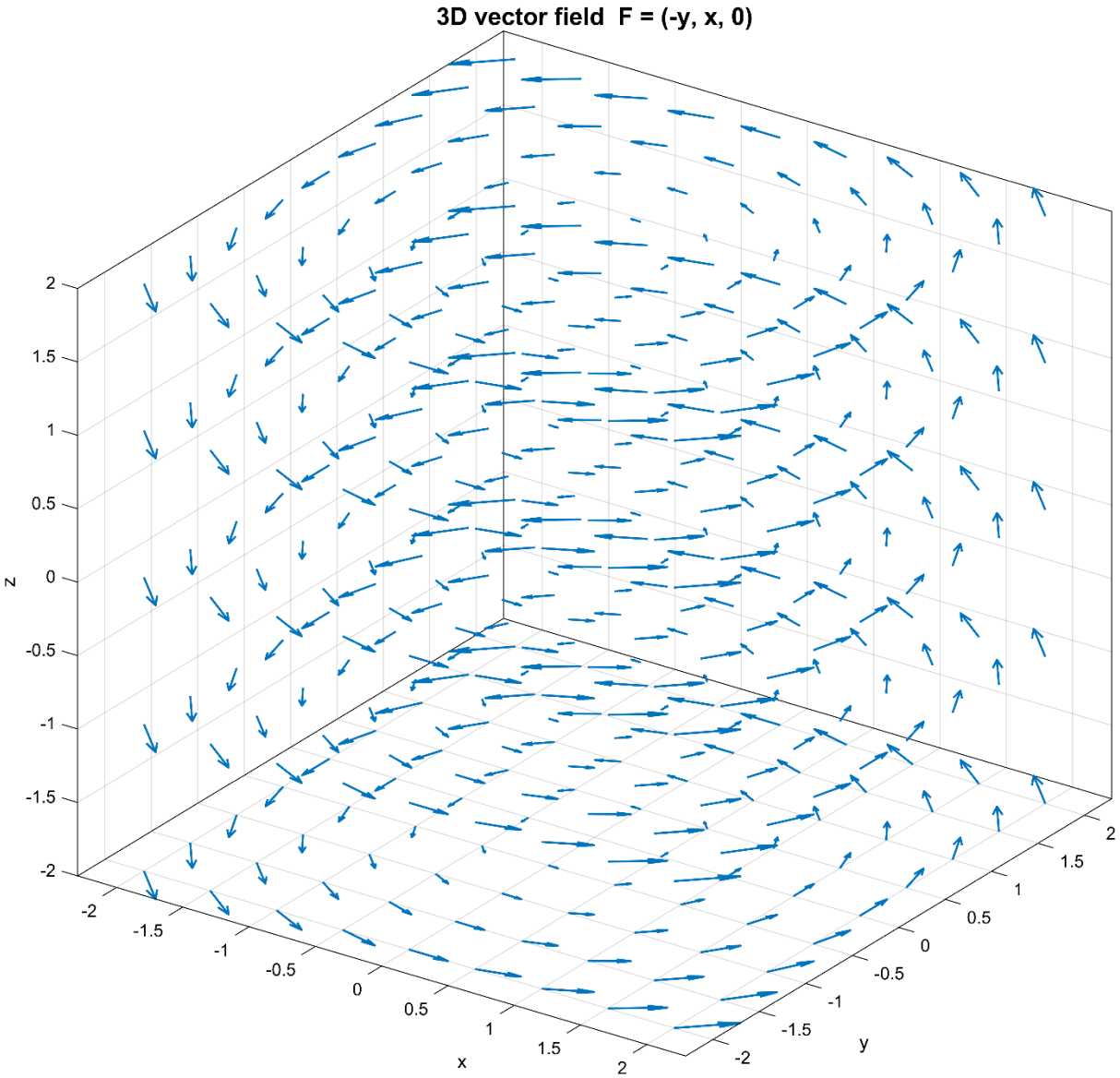


Figure 1. Three-dimensional view of the rotational field $\mathbf{F} = (-y, x, 0)$. The arrows circulate counter-clockwise around the z-axis and grow longer with the distance from it.

3.2 Divergence and curl

The divergence is obtained by differentiating each component with respect to its own coordinate:

$$\nabla\cdot\mathbf{F} = \partial(-y)/\partial x + \partial(x)/\partial y + \partial(0)/\partial z = 0 \tag{11}$$

and the curl has only a z-component:

$$\nabla\times\mathbf{F} = \mathbf{a}_z[\partial x/\partial x - \partial(-y)/\partial y] = \mathbf{a}_z(1 + 1) = 2\mathbf{a}_z \tag{12}$$

The field therefore has no sources or sinks, but it has a constant, non-zero curl along +z. The paddle-wheel interpretation is consistent with the physical picture: a wheel placed anywhere in the field spins about the z-axis with angular velocity $\tfrac{1}{2}|\nabla\times\mathbf{F}| = 1$ rad/s, which is exactly the angular velocity of the rotation that generates the field. In general, $\nabla\times(\boldsymbol{\omega}\times\mathbf{r}) = 2\boldsymbol{\omega}$.

3.3 Behaviour on the three planes

The slicing planes of Figure 2 are $z = 0$, $y = 1$ and $x = 1$. Table 2 summarises what appears on each of them.

Table 2. Field and curl components on the three planes for F = (−y, x, 0).

Plane	In-plane vector	Out-of-plane component	Curl component	Observation
XY (z = 0)	$(-y, x)$	$F_z = 0$	$(\nabla\times\mathbf{F})_z = 2$	Counter-clockwise circulation
XZ (y = 1)	$(-1, 0)$	$F_y = x$	$(\nabla\times\mathbf{F})_y = 0$	Uniform field, no circulation
YZ (x = 1)	$(1, 0)$	$F_x = -y$	$(\nabla\times\mathbf{F})_x = 0$	Uniform field, no circulation

On the XY plane the circulation is evident. On the XZ and YZ planes the in-plane vectors are uniform arrows, which is a first hint that nothing rotates in those planes; the rotation is carried entirely by the component normal to each plane ($F_y = x$ and $F_x = -y$), which the 2D plot cannot display. This is exactly the situation described by the remark in Section 2.3.

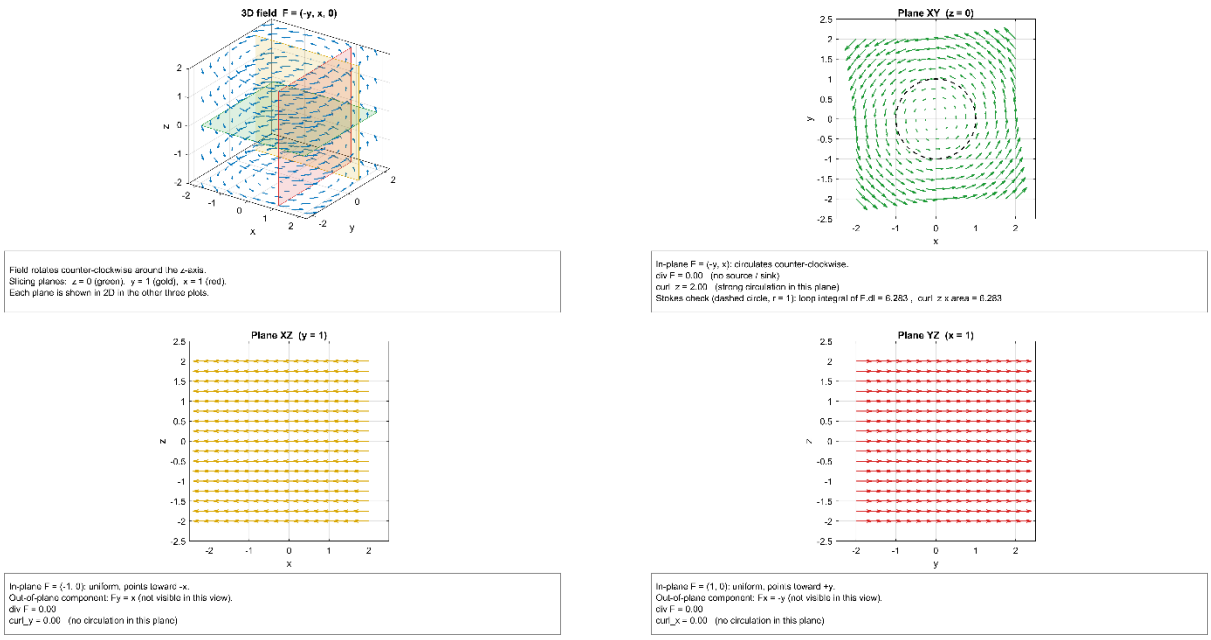


Figure 2. The field $\mathbf{F} = (-y, x, 0)$ with the slicing planes $z = 0$ (green), $y = 1$ (yellow) and $x = 1$ (red), and its 2D views on each plane. The dashed circle on the XY plot is the Stokes contour of radius 1.

3.4 Verification with Stokes' theorem

The circulation of $\mathbf{F}$ along the unit circle in the XY plane, $x = \cos\theta$, $y = \sin\theta$, is

$$\oint \mathbf{F}\cdot d\mathbf{l} = \int_0^{2\pi} (\sin^2\theta + \cos^2\theta)\, d\theta = 2\pi \approx 6.2832 \qquad (13)$$

while the surface integral of the curl over the enclosed disc is $(\nabla\times\mathbf{F})_z \cdot \pi r^2 = 2 \cdot \pi = 2\pi$. The two values agree, and the MATLAB output (*Circulation on r = 1 circle = 6.2832; curl_z × area = 6.2832*) confirms it numerically. Figure 3 shows the field next to its curl, which is a uniform vector of length 2 pointing along +z at every grid point, together with the summary of results.

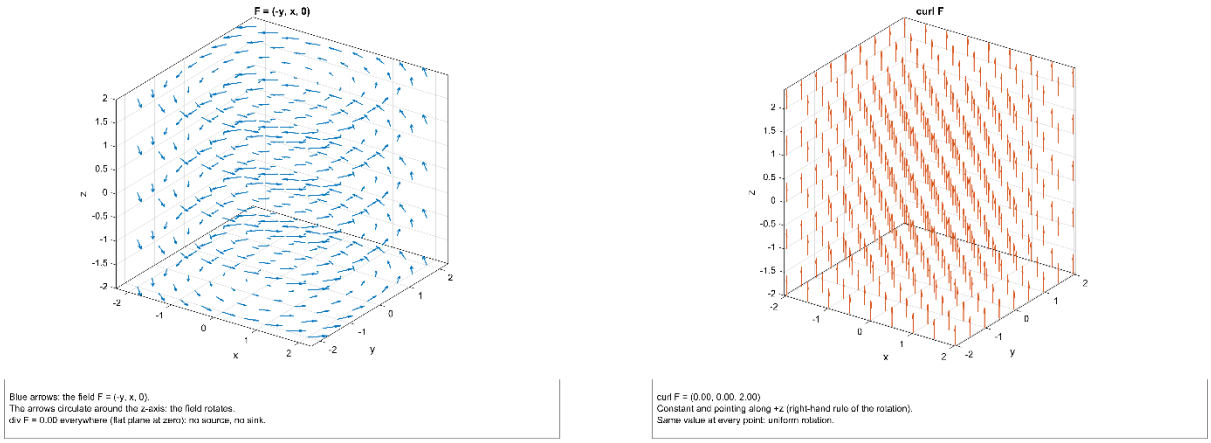


Figure 3. The field $\mathbf{F} = (-y, x, 0)$ (left) and its curl $\nabla\times\mathbf{F} = (0, 0, 2)$ (right). The curl is constant and uniform, which reflects the uniform rotation of the field.

> **Key result.** For $\mathbf{F} = (-y, x, 0)$: $\nabla\cdot\mathbf{F} = 0$ and $\nabla\times\mathbf{F} = 2\mathbf{a}_z$. The field is solenoidal but not irrotational, and the circulation around the unit circle, 2π, equals the flux of the curl through the disc, as required by Stokes' theorem.

4. Case Study 2: A Field with Zero Divergence and Zero Curl

4.1 Description of the field

The second field is a saddle (hyperbolic) field:

$$\mathbf{F} = x\,\mathbf{a}_x - y\,\mathbf{a}_y + 0\,\mathbf{a}_z \tag{14}$$

Its magnitude is again $|\mathbf{F}| = r$, the same as in Case 1, but the behaviour is completely different, which shows that the magnitude alone says nothing about the divergence or the curl. Along the x-axis the field points away from the origin; along the y-axis it points toward the origin. The origin is a stagnation point where $\mathbf{F} = 0$. The streamlines satisfy $dx/x = dy/(-y)$, that is, $xy = C$: rectangular hyperbolas. The field is shown in Figure 4.

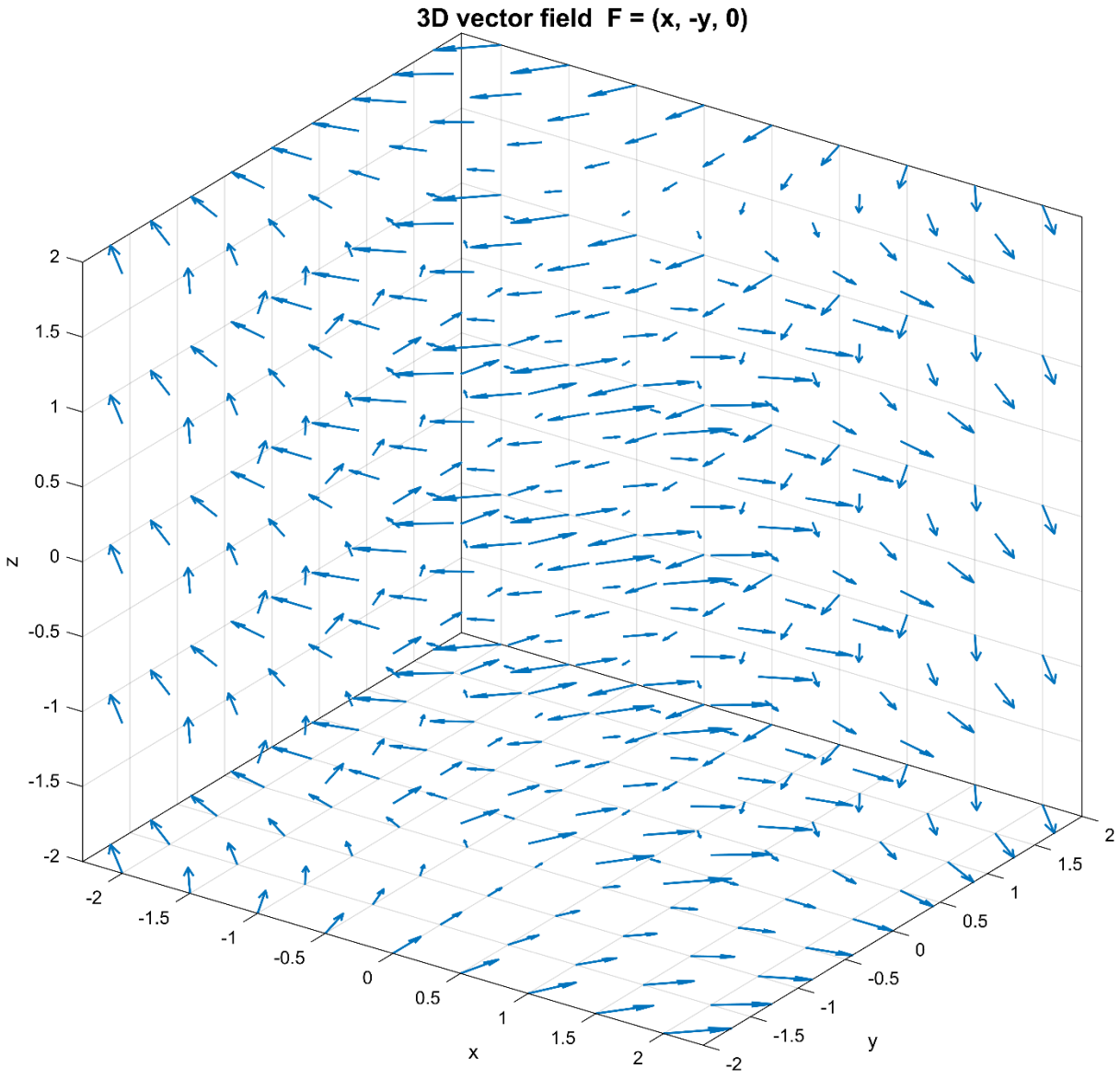


Figure 4. Three-dimensional view of the saddle field $\mathbf{F} = (x, -y, 0)$.

4.2 Divergence and curl

$$\nabla\cdot\mathbf{F} = \partial x/\partial x + \partial(-y)/\partial y + \partial(0)/\partial z = 1 - 1 + 0 = 0 \tag{15}$$

$$\nabla\times\mathbf{F} = \mathbf{a}_x(0-0) + \mathbf{a}_y(0-0) + \mathbf{a}_z[\partial(-y)/\partial x - \partial x/\partial y] = 0 \tag{16}$$

The expansion in the x-direction ($\partial F_x/\partial x = +1$) is exactly cancelled by the compression in the y-direction ($\partial F_y/\partial y = -1$), so the field is solenoidal. Since $\partial F_y/\partial x$ and $\partial F_x/\partial y$ are both zero, there is no rotation, and the field is irrotational.

4.3 Scalar potential

Because $\nabla\times\mathbf{F} = 0$, the field derives from a scalar potential. Integrating $F_x = \partial\varphi/\partial x = x$ and $F_y = \partial\varphi/\partial y = -y$ gives

$$\varphi(x, y) = (x^2 - y^2)/2, \quad \mathbf{F} = \nabla\varphi \tag{17}$$

The potential satisfies Laplace's equation, $\nabla^2\varphi = 1 - 1 = 0$, which is the same statement as $\nabla\cdot\mathbf{F} = 0$ for a gradient field. The equipotential lines $x^2 - y^2 = C$ are hyperbolas orthogonal to the streamlines $xy = C$. Fields of this type appear in electrostatics near the centre of a hyperbolic-electrode quadrupole, and in fluid mechanics near a stagnation point.

4.4 Behaviour on the three planes

Table 3. Field and curl components on the three planes for F = (x, −y, 0).

Plane	In-plane vector	Out-of-plane component	Curl component	Observation
XY (z = 0)	$(x, -y)$	$F_z = 0$	$(\nabla\times\mathbf{F})_z = 0$	Saddle flow, no circulation
XZ (y = 1)	$(x, 0)$	$F_y = -1$	$(\nabla\times\mathbf{F})_y = 0$	Horizontal arrows pointing away from x = 0
YZ (x = 1)	$(-y, 0)$	$F_x = 1$	$(\nabla\times\mathbf{F})_x = 0$	Horizontal arrows pointing toward y = 0

On the XY plane the saddle pattern is clearly visible. The XZ and YZ views show horizontal arrows only, because $F_z = 0$; the constant components $F_y = -1$ and $F_x = 1$ normal to those planes are stated in the annotations of Figure 5.

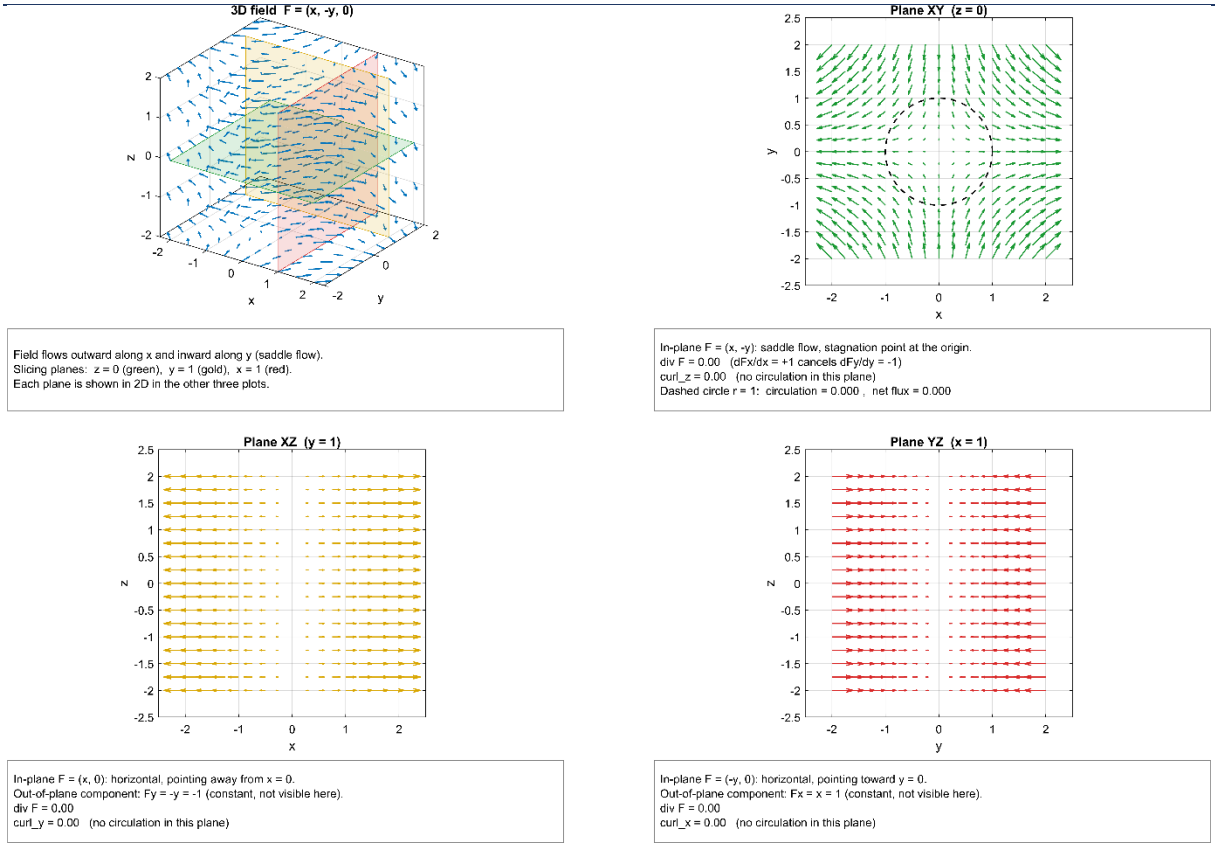


Figure 5. The field $\mathbf{F} = (x, -y, 0)$ with the planes $z = 0$, $y = 1$ and $x = 1$, and its 2D views. The dashed circle on the XY plot is the contour used for the circulation and flux checks.

4.5 Divergence and curl maps

Since the divergence and the curl are zero everywhere, an arrow plot of the curl would be empty. To keep the presentation informative, each grid point is drawn as a dot coloured by the local value of the quantity on a fixed scale. In Figure 6 the colour represents $\nabla \cdot \mathbf{F}$ on a symmetric scale from −1 (blue, sink) to +1 (red, source); every dot has the neutral grey of the zero value, so no sources and no sinks exist. In Figure 7 the colour represents $|\nabla \times \mathbf{F}|$ on a scale from 0 to 1, and again every dot is neutral: the field is irrotational.

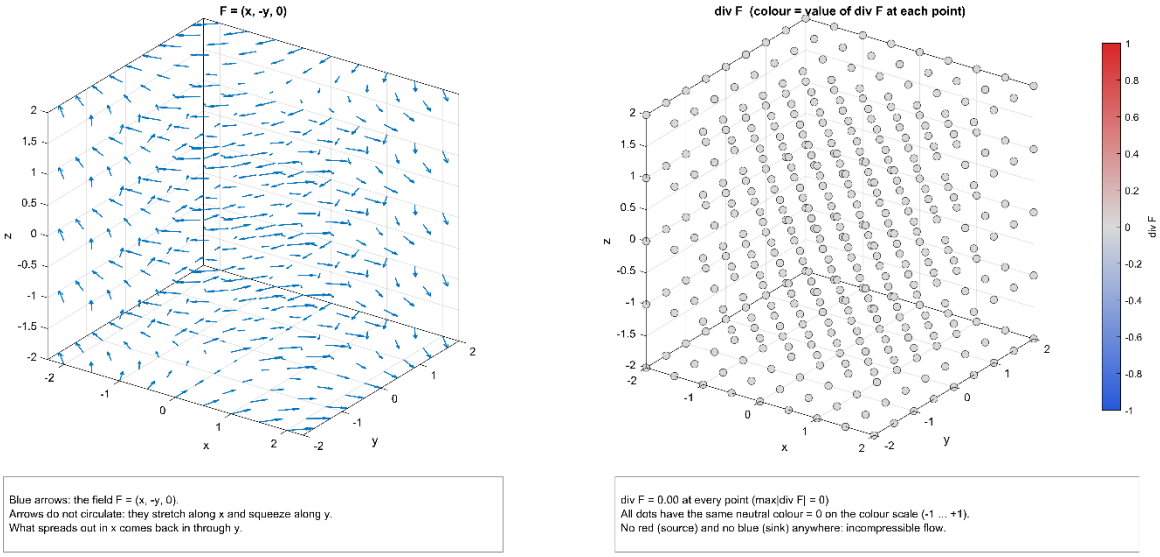


Figure 6. The field $\mathbf{F} = (x, -y, 0)$ (left) and the colour map of $\nabla\cdot\mathbf{F}$ (right). All points are neutral, i.e. $\nabla\cdot\mathbf{F} = 0$ everywhere.

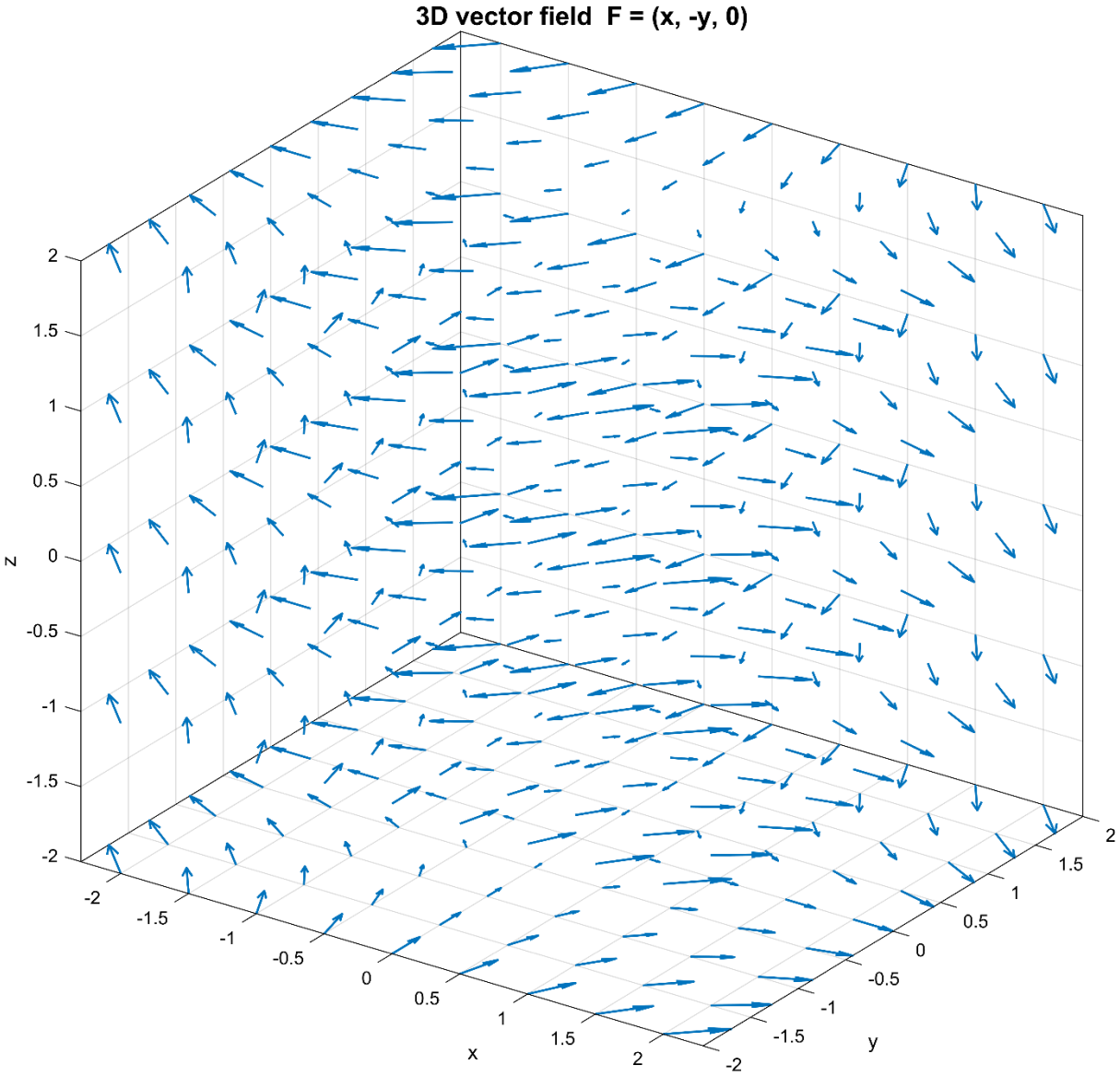


Figure 7. Colour map of $|\nabla\times\mathbf{F}|$ for $\mathbf{F} = (x, -y, 0)$. The curl is the zero vector at every grid point.

4.6 Verification of the theorems

On the unit circle, with $d\mathbf{l} = (-\sin\theta, \cos\theta)\, d\theta$ and outward normal $\mathbf{n} = (\cos\theta, \sin\theta)$, the circulation and the net flux are

$$\oint \mathbf{F}\cdot d\mathbf{l} = \int_0^{2\pi} (-2\sin\theta\cos\theta)\, d\theta = 0 \qquad (18)$$
$$\oint \mathbf{F}\cdot\mathbf{n}\, dl = \int_0^{2\pi} (\cos^2\theta - \sin^2\theta)\, d\theta = 0$$

Both vanish, in agreement with Stokes' theorem (zero curl) and with the divergence theorem (zero divergence). The program output confirms this numerically (circulation = 0.0000 and flux = 0.0000).

4.7 Comparison of the two fields

Table 4. Comparison of the two test fields.

Property	Case 1: $\mathbf{F} = (-y, x, 0)$	Case 2: $\mathbf{F} = (x, -y, 0)$
Magnitude $\lvert\mathbf{F}\rvert$	r	r
Divergence $\nabla\cdot\mathbf{F}$	0	0
Curl $\nabla\times\mathbf{F}$	$2\mathbf{a}_z$	0
Scalar potential	does not exist (not conservative)	$\varphi = (x^2 - y^2)/2$
Flow pattern	rigid rotation	saddle (stagnation point)
Circulation, r = 1	2π	0
Net flux, r = 1	0	0

Key result. For $\mathbf{F} = (x, -y, 0)$: $\nabla\cdot\mathbf{F} = 0$ and $\nabla\times\mathbf{F} = 0$. The field is both solenoidal and irrotational, it derives from the harmonic potential $\varphi = (x^2 - y^2)/2$, and both the circulation and the net flux on any closed contour vanish.

5. Uniform Plane Electromagnetic Wave

5.1 Maxwell's equations in free space

In a source-free region of free space ($\rho_v = 0$, $\mathbf{J} = 0$) Maxwell's equations reduce to

$$\begin{aligned} \nabla\cdot\mathbf{E} &= 0 \\ \nabla\cdot\mathbf{H} &= 0 \\ \nabla\times\mathbf{E} &= -\mu_0\,\partial\mathbf{H}/\partial t \\ \nabla\times\mathbf{H} &= \varepsilon_0\,\partial\mathbf{E}/\partial t \end{aligned} \tag{19}$$

These four equations are built from exactly the operators studied in Sections 2 to 4. The two divergence equations state that neither field has sources, and the two curl equations couple **E** and **H**: a time-varying **H** creates a circulating **E** (Faraday's law), and a time-varying **E** creates a circulating **H** (Ampère's law with displacement current). This mutual coupling is what allows a wave to propagate.

5.2 Wave equation

Taking the curl of Faraday's law, using $\nabla\times(\nabla\times\mathbf{E}) = \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E}$ and $\nabla\cdot\mathbf{E} = 0$, and substituting Ampère's law gives the wave equation

$$\nabla^2\mathbf{E} = \mu_0\varepsilon_0\,\partial^2\mathbf{E}/\partial t^2 \tag{20}$$

which describes waves with speed $c = 1/\sqrt{(\mu_0\varepsilon_0)} \approx 3 \times 10^8$ m/s. The ratio of the electric to the magnetic field of such a wave is the intrinsic impedance of free space:

$$\eta_0 = \sqrt{(\mu_0/\varepsilon_0)} \approx 120\pi \approx 377\ \Omega \tag{21}$$

5.3 The plane wave in a general direction

A uniform plane wave travelling in the direction of the unit vector **n** is described by

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= E_0 \cos\xi\,\mathbf{p} \\ \mathbf{H}(\mathbf{r}, t) &= (1/\eta)\,\mathbf{n}\times\mathbf{E} = (E_0/\eta)\cos\xi\,\mathbf{h} \end{aligned} \tag{22}$$

with the phase

$$\xi = k\,\mathbf{n}\cdot\mathbf{r} - \omega t, \quad k = \omega/c = 2\pi/\lambda \tag{23}$$

Here **p** is the unit vector along **E** (the polarisation), $\mathbf{h} = \mathbf{n}\times\mathbf{p}$ is the unit vector along **H**, and $H_0 = E_0/\eta$. The condition $\nabla\cdot\mathbf{E} = 0$ forces $\mathbf{n}\cdot\mathbf{p} = 0$: the wave is transverse. The triad (**p**, **h**, **n**) is orthonormal and right-handed, $\mathbf{p}\times\mathbf{h} = \mathbf{n}$. The surfaces of constant phase, $\mathbf{n}\cdot\mathbf{r}$ = constant, are planes perpendicular to **n** (the wavefronts); on each of them the fields are uniform, which is what “uniform plane wave” means. All lengths in the figures are given in wavelengths, so that $\lambda = 1$ and $k = 2\pi$.

5.4 Direction of propagation and polarisation

To obtain a completely general direction, **n** is defined by the polar angle θ (from the +z axis) and the azimuth φ (from the +x axis). Two unit vectors perpendicular to **n** and to each other, $\mathbf{e}_1$ and $\mathbf{e}_2$, span the transverse plane, and the polarisation angle ψ fixes **p** inside it:

$$\begin{aligned} \mathbf{n} &= (\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta) \\ \mathbf{e}_1 &= (\cos\theta\cos\varphi,\ \cos\theta\sin\varphi,\ -\sin\theta) \\ \mathbf{e}_2 &= (-\sin\varphi,\ \cos\varphi,\ 0) \\ \mathbf{p} &= \cos\psi\,\mathbf{e}_1 + \sin\psi\,\mathbf{e}_2, \quad \mathbf{h} = \mathbf{n}\times\mathbf{p} \end{aligned} \tag{24}$$

The values used in the simulation are listed in Table 5, and the resulting vectors are given in Table 6. The check $\mathbf{n}\cdot\mathbf{p} = 0$ is satisfied to machine precision.

Table 5. Simulation parameters of the plane wave.

Parameter	Symbol	Value
Polar angle of n	θ	50°
Azimuth of n	φ	35°
Polarisation angle	ψ	30°
Electric field amplitude	E_0	1 V/m
Intrinsic impedance	η	$120\pi \approx 377\ \Omega$
Magnetic field amplitude	$H_0 = E_0/\eta$	≈ 2.65 mA/m
Wavenumber (lengths in wavelengths)	k	2π
Snapshot phase	ωt	0 rad
Plotted cube	—	±0.75 λ on each axis

Table 6. Direction vectors of the simulated wave.

Vector	Meaning	Components (x, y, z)
n	propagation direction	(0.628, 0.439, 0.643)
p	direction of **E**	(0.169, 0.729, −0.663)
h = n × p	direction of **H**	(−0.760, 0.525, 0.383)

5.5 Divergence, curl and gradient of the wave

For a field of the form $\mathbf{A}\,g(\xi)$ with a constant vector **A**, and since $\nabla\xi = k\mathbf{n}$, the chain rule gives

$$\begin{aligned} \nabla\cdot[\mathbf{A}\,g(\xi)] &= k\,(\mathbf{n}\cdot\mathbf{A})\,g'(\xi) \\ \nabla\times[\mathbf{A}\,g(\xi)] &= k\,\mathbf{n}\times\mathbf{A}\,g'(\xi) \\ \nabla\psi(\xi) &= k\,\mathbf{n}\,\psi'(\xi) \end{aligned} \tag{25}$$

Applying these identities to **E** and **H** gives the eight quantities of Table 7. Two comments are useful. First, $\nabla\cdot\mathbf{E} = \nabla\cdot\mathbf{H} = 0$ because **n** is perpendicular to both fields. Second, the curls are shifted by 90° with respect to the fields (sin ξ instead of cos ξ): where the field is maximum its

curl is zero, and where the field crosses zero its curl is maximum, with amplitude kE_0 and kH_0 respectively.

Table 7. Analytical expressions for the wave fields ($\xi = \mathrm{k}\,\mathbf{n}\cdot\mathbf{r} - \omega t$).

Quantity	Expression	Direction	Comment
E	$E_0 \cos\xi$	**p**	in phase with **H**
∇×E	$-kE_0 \sin\xi$	**h**	Faraday's law
∇·E	0	—	$\mathbf{n}\cdot\mathbf{p} = 0$
∇|E|	$-kE_0\,\mathrm{sgn}(\cos\xi)\sin\xi$	**n**	$\lvert\mathbf{E}\rvert = E_0\lvert\cos\xi\rvert$
H	$H_0 \cos\xi$	**h**	$H_0 = E_0/\eta$
∇×H	$+kH_0 \sin\xi$	**p**	Ampère's law
∇·H	0	—	$\mathbf{n}\cdot\mathbf{h} = 0$
∇|H|	$-kH_0\,\mathrm{sgn}(\cos\xi)\sin\xi$	**n**	$\lvert\mathbf{H}\rvert = H_0\lvert\cos\xi\rvert$

The two curl expressions are consistent with Maxwell's equations. Since $\mu_0\omega/\eta_0 = \omega\sqrt{(\mu_0\varepsilon_0)} = k$,

$$-\mu_0\,\partial\mathbf{H}/\partial t = -\mu_0\omega H_0 \sin\xi\,\mathbf{h} = -kE_0 \sin\xi\,\mathbf{h} = \nabla\times\mathbf{E} \tag{26}$$

and, since $\varepsilon_0\omega\eta_0 = k$,

$$\varepsilon_0\,\partial\mathbf{E}/\partial t = \varepsilon_0\omega E_0 \sin\xi\,\mathbf{p} = kH_0 \sin\xi\,\mathbf{p} = \nabla\times\mathbf{H} \tag{27}$$

The magnitudes $|\mathbf{E}| = E_0|\cos\xi|$ and $|\mathbf{H}| = H_0|\cos\xi|$ of the instantaneous fields depend only on the phase, so their gradients are parallel to **n** everywhere. They vanish at the crests and troughs of the wave (where $\sin\xi = 0$) and reach the value kE_0 (respectively kH_0) at the zero crossings of the field, where $|\mathbf{E}|$ has a kink and the gradient changes sign.

Finally, the Poynting vector gives the direction and density of power flow:

$$\mathbf{S} = \mathbf{E}\times\mathbf{H} = (E_0^2/\eta)\cos^2\xi\,\mathbf{n}, \quad S_{\mathrm{av}} = E_0^2/(2\eta) \tag{28}$$

It always points along +**n**, and for $E_0 = 1$ V/m its time average is $1/(2\times 377) \approx 1.33$ mW/m².

5.6 Visualisation of E and H and their operators

Each of the eight quantities of Table 7 is shown in its own figure containing four graphs. The first graph is the three-dimensional view: a cube of side 1.5 λ sampled with arrows (or coloured dots for the divergence), the three coordinate planes drawn as translucent squares, and a black arrow indicating the direction of propagation **n**. The other three graphs show the quantity on the XY ($z = 0$), XZ ($y = 0$) and YZ ($x = 0$) planes through the origin. For the vector quantities the background colour gives the magnitude and the arrows give the in-plane projection of the vector; for the divergence the colour gives the value of ∇·**E** or ∇·**H** on a fixed symmetric scale, so that a zero value is immediately recognisable.

5.6.1 The electric field E

In Figure 8 the arrows are parallel to **p** and their length varies as cos ξ. They are identical on every plane perpendicular to **n**, and they reverse direction every half wavelength along **n**. On the plane maps this appears as parallel bands, perpendicular to the projection of **n** on that plane; the in-plane arrows are short on a plane that is nearly perpendicular to **p**.

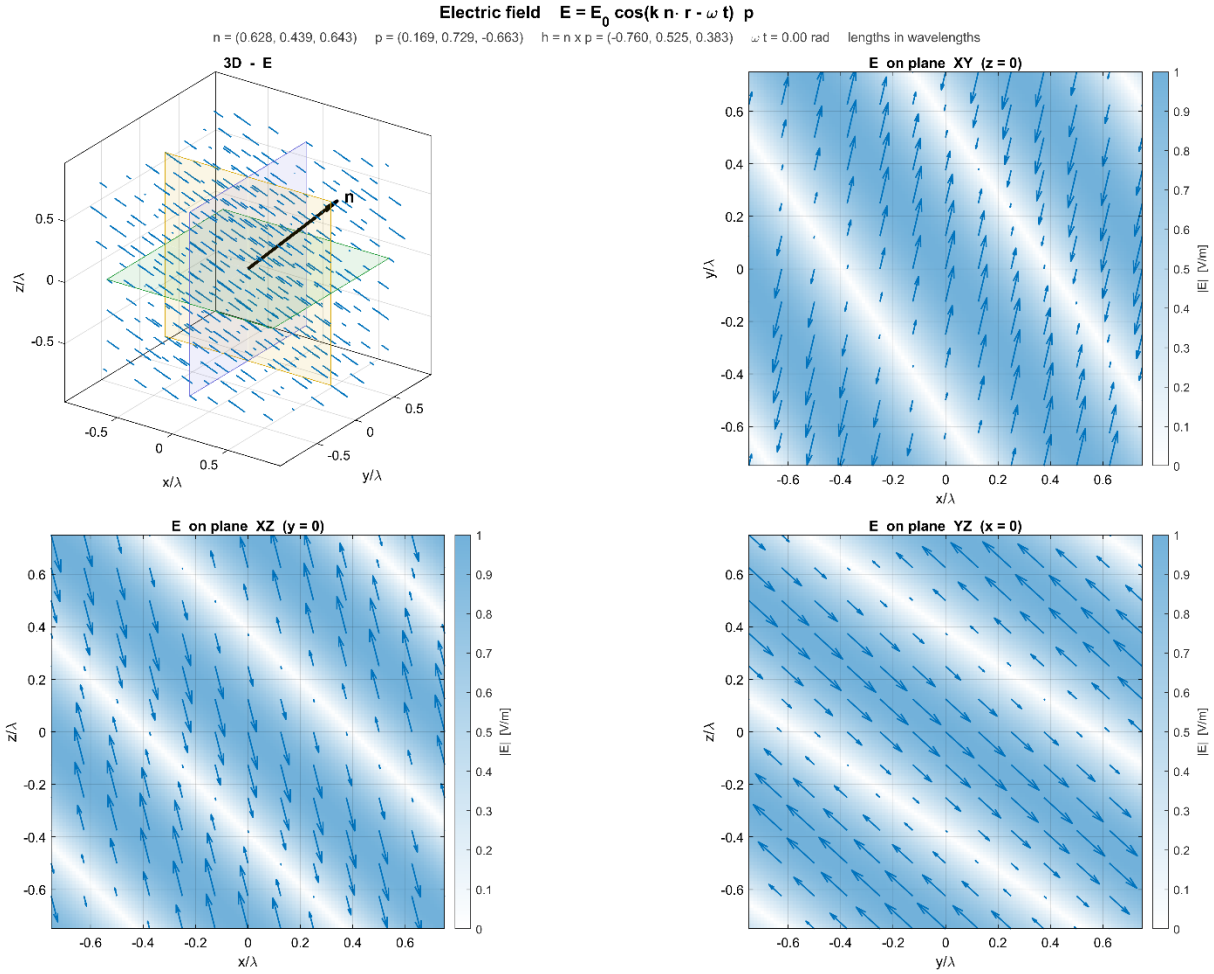


Figure 8. The electric field **E** = E_0 cos ξ **p** in 3D and on the three coordinate planes.

Figure 9 shows $\nabla\times\mathbf{E} = -kE_0 \sin\xi\,\mathbf{h}$. The curl has the same banded structure as the field, but it points along **h** (the direction of **H**), as Faraday's law requires, and it is shifted by a quarter of a period in space with respect to **E**. Its amplitude is 2π times larger than E_0 because the derivatives are taken per wavelength.

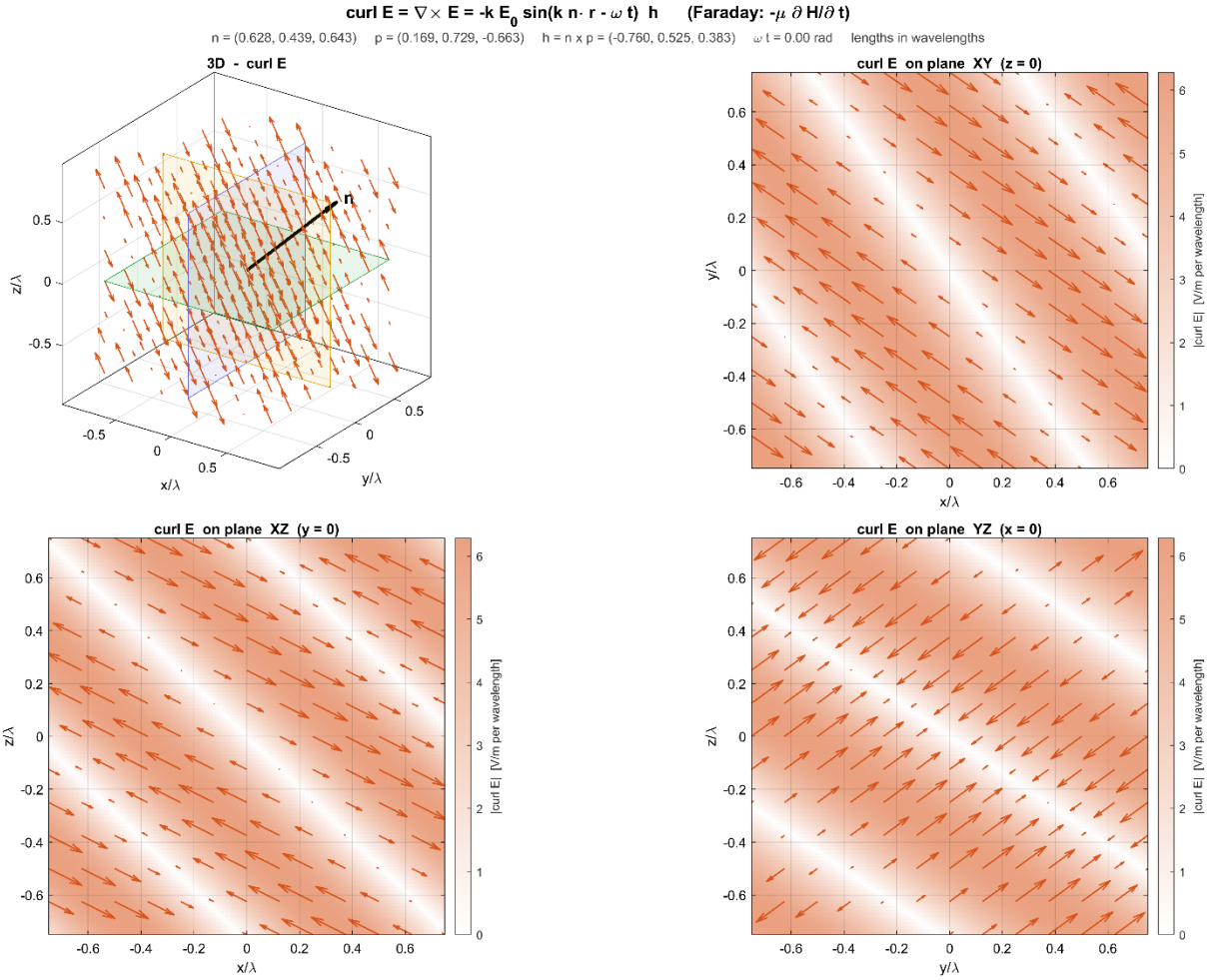


Figure 9. The curl of the electric field, $\nabla\times\mathbf{E} = -kE_0 \sin\xi\,\mathbf{h}$, in 3D and on the three coordinate planes.

In Figure 10 the colour map of $\nabla\cdot\mathbf{E}$ is uniformly neutral on the scale from $-kE_0$ to $+kE_0$. A longitudinal component of **E** along **n** would have produced alternating red and blue bands; their absence confirms that the wave is transverse and that the electric field has no sources.

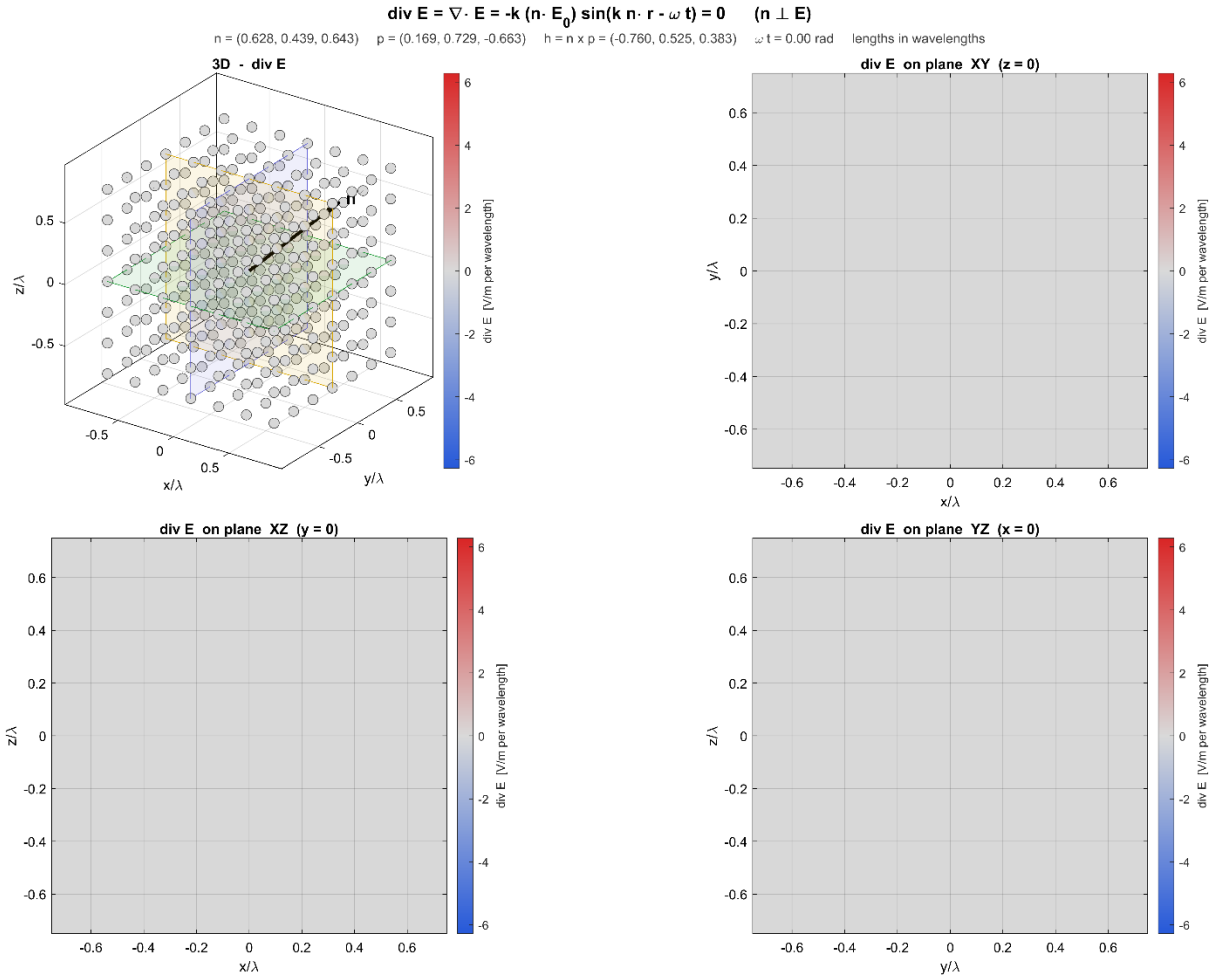


Figure 10. The divergence of the electric field. Every point has $\nabla\cdot\mathbf{E} = 0$ because $\mathbf{n}\cdot\mathbf{p} = 0$.

Figure 11 shows ∇|**E**|. The gradient is parallel to **n** on all planes, so the arrows in the 2D views are the projection of **n** and are all parallel; their sign alternates from one half-wavelength band to the next and they vanish where |**E**| is at a maximum.

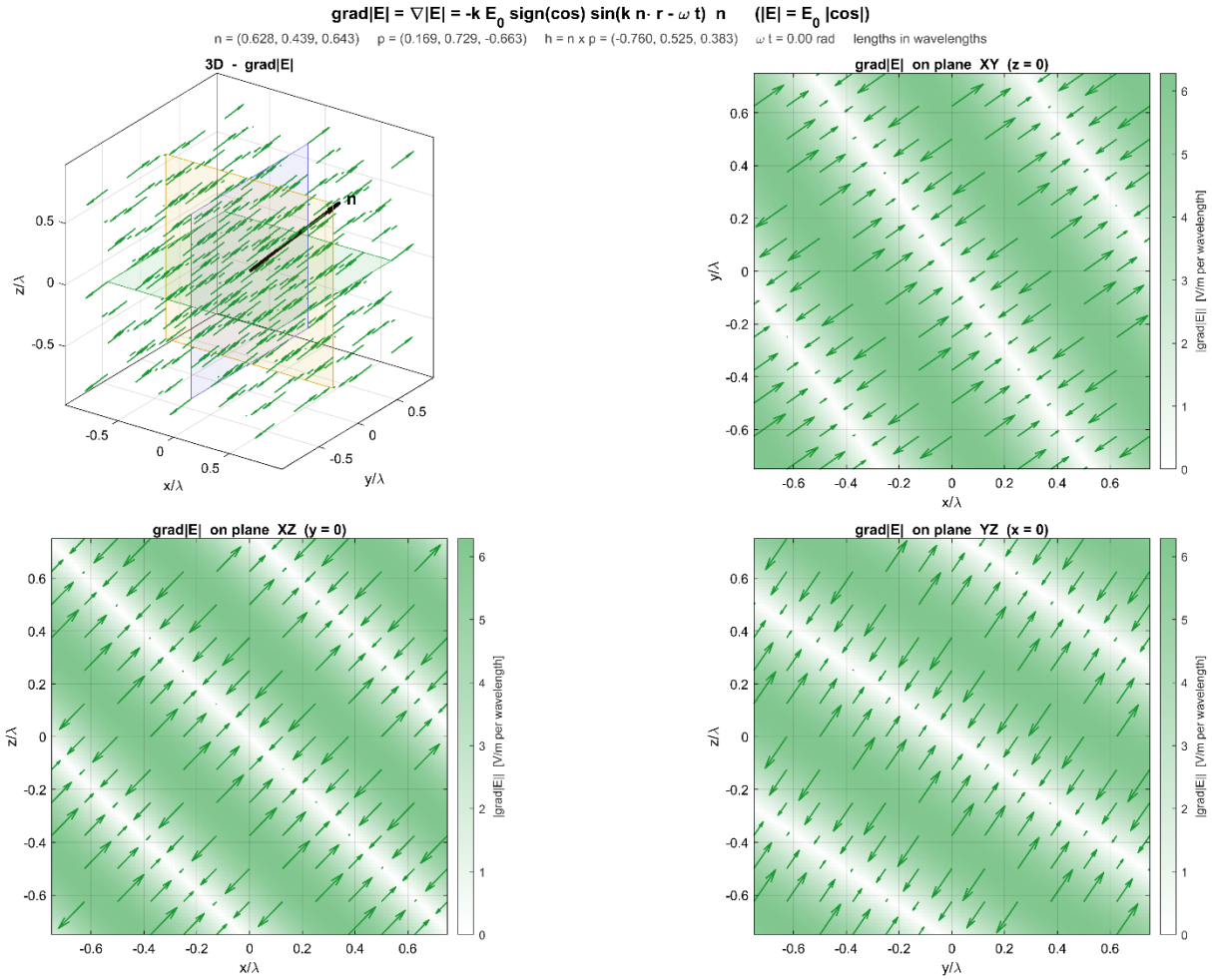


Figure 11. The gradient of the magnitude of the electric field, ∇|**E**|, in 3D and on the three coordinate planes.

5.6.2 The magnetic field H

The magnetic field (Figure 12) has the same spatial pattern as **E**, but it points along **h**, perpendicular to **E** and to **n**, and its amplitude is smaller by the factor $1/\eta \approx 1/377$. The two fields are in phase: they reach their maxima and their zeros at the same places.

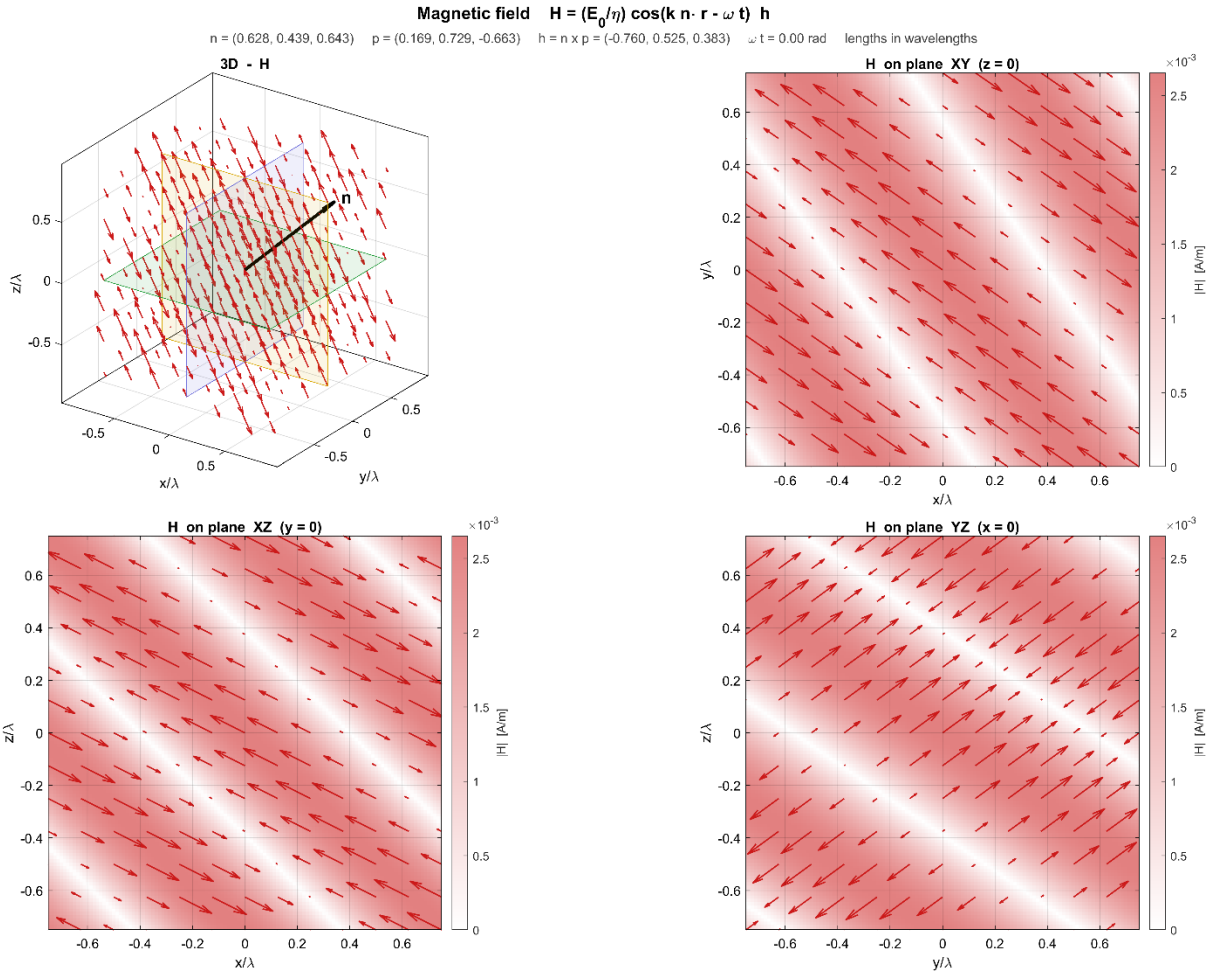


Figure 12. The magnetic field **H** = (E_0/η) cos ξ **h** in 3D and on the three coordinate planes.

Figure 13 shows $\nabla\times\mathbf{H} = +kH_0 \sin\xi\,\mathbf{p}$. The curl of **H** is parallel to **E**, as Ampère's law requires, and it is again in quadrature with the field that generates it. Notice the opposite sign compared with the curl of **E**, which follows from $\mathbf{n}\times\mathbf{h} = -\mathbf{p}$.

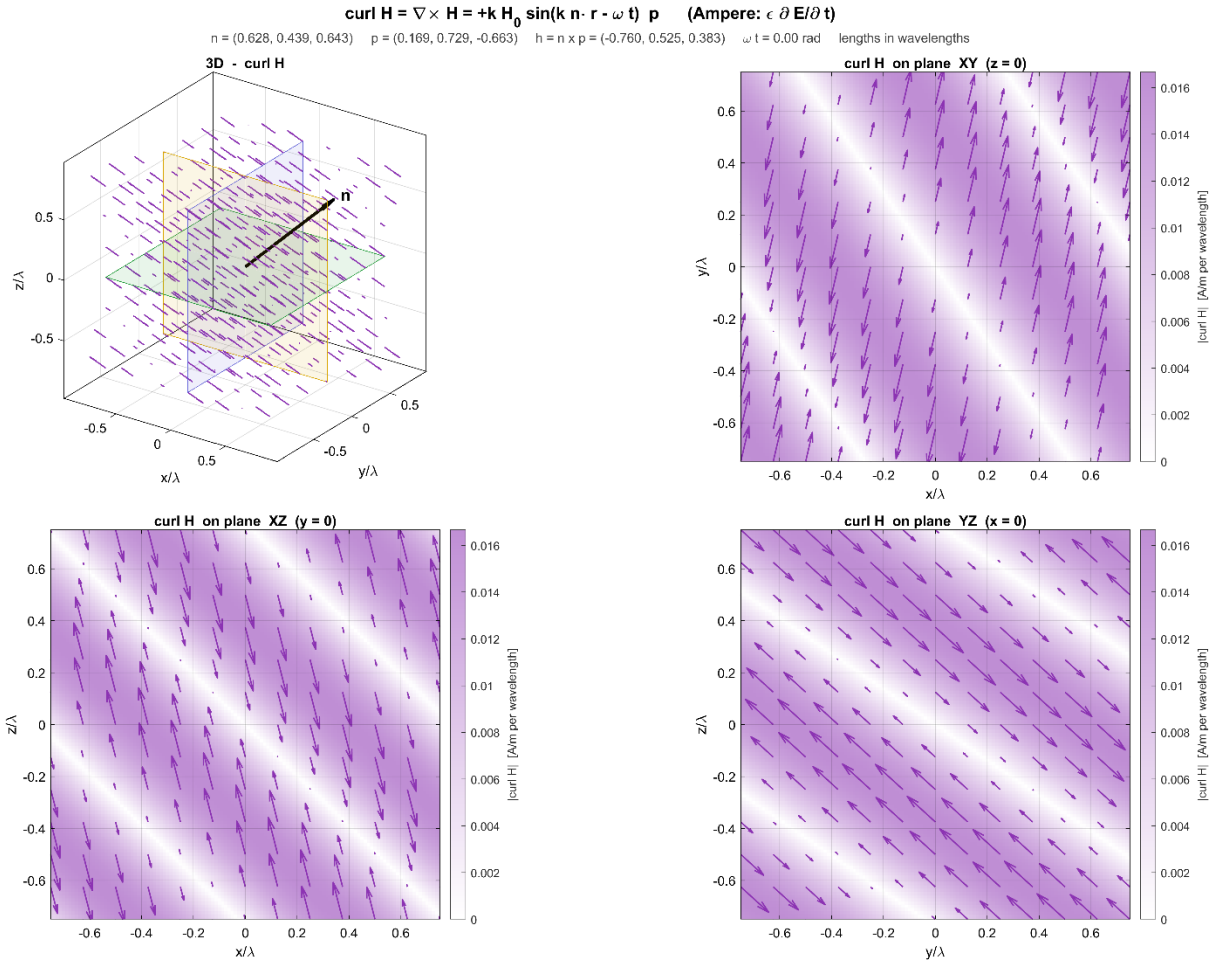


Figure 13. The curl of the magnetic field, $\nabla\times\mathbf{H} = kH_0 \sin\xi\,\mathbf{p}$, in 3D and on the three coordinate planes.

As for the electric field, the divergence of **H** (Figure 14) is zero at every point: magnetic field lines have no beginning or end, and the wave carries no longitudinal magnetic component.

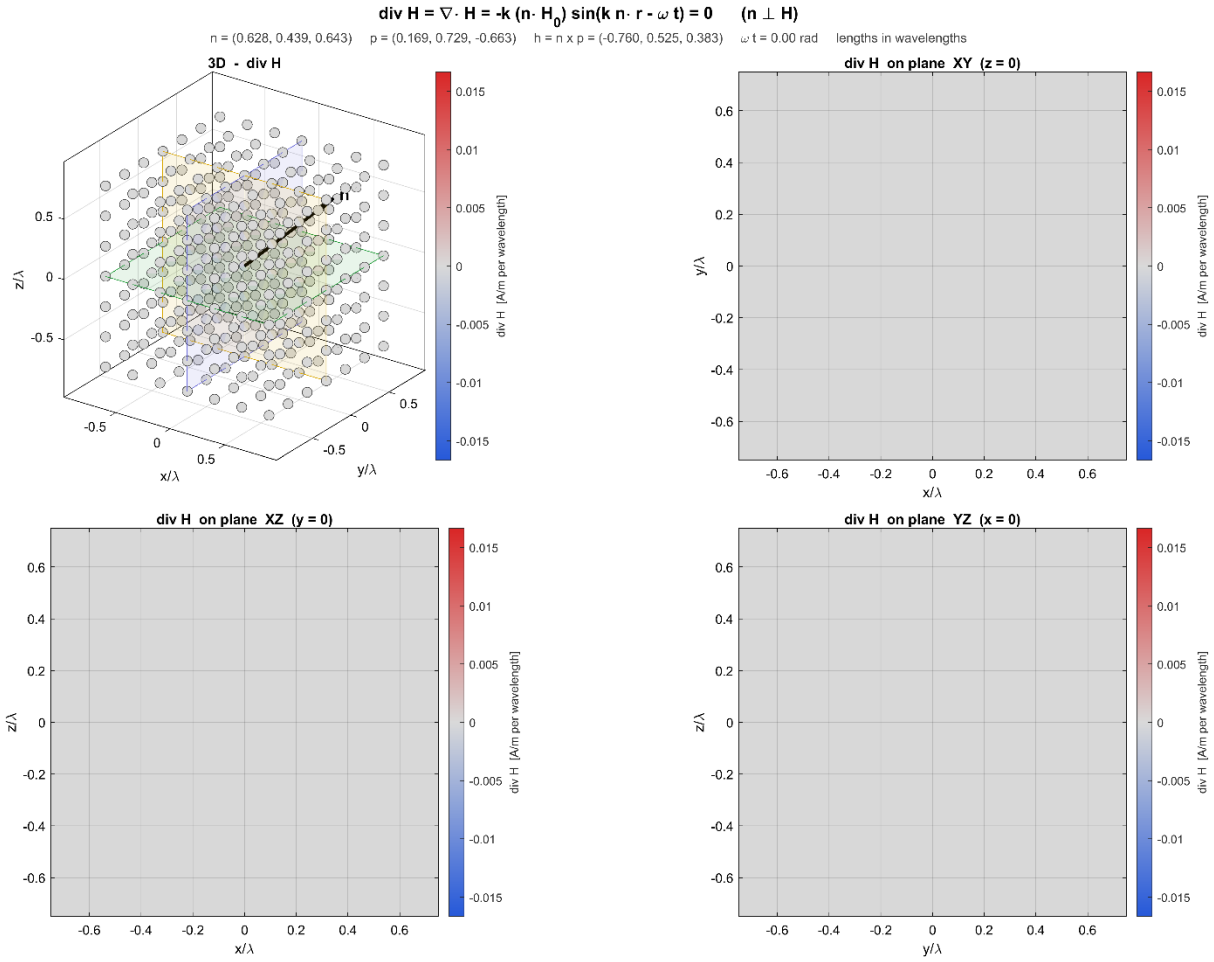


Figure 14. The divergence of the magnetic field. Every point has $\nabla \cdot \mathbf{H} = 0$ because $\mathbf{n} \cdot \mathbf{h} = 0$.

Figure 15 shows $\nabla|\mathbf{H}|$, which has the same structure as $\nabla|\mathbf{E}|$ (parallel to **n**, alternating sign) scaled by $1/\eta$.

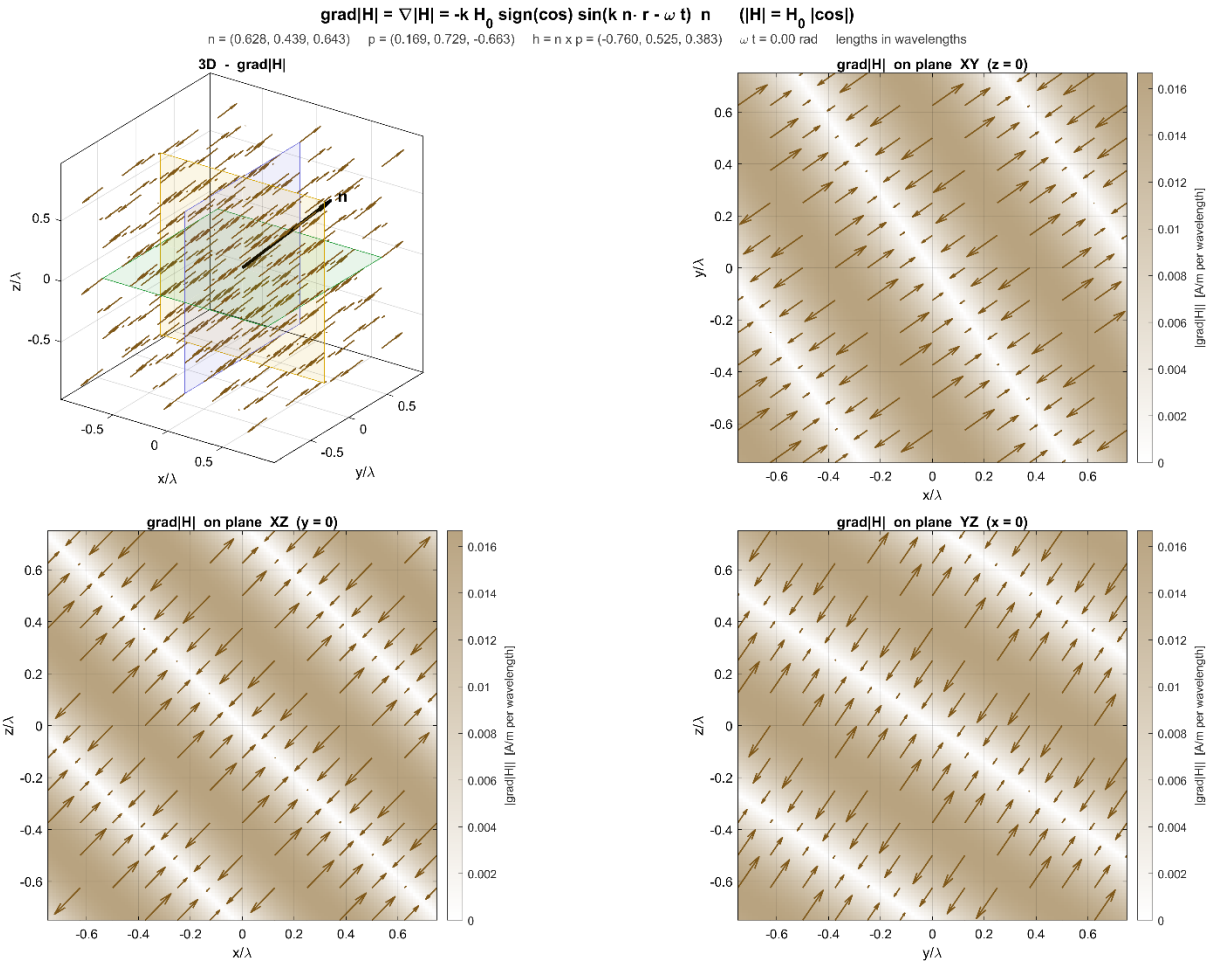


Figure 15. The gradient of the magnitude of the magnetic field, $\nabla|\mathbf{H}|$, in 3D and on the three coordinate planes.

5.7 The wave in three dimensions

Figure 16 draws **E** (blue) and **H** (red) along the axis of propagation. The black line is the axis through the origin along **n**, the two sinusoidal curves are the tips of the field vectors, the thin arrows are the field vectors at regularly spaced points, and the translucent square is one wavefront. The black arrow at the end of the axis is the Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$. For readability **E** and **H** are drawn with the same length (**H** multiplied by η). The figure shows that **E** and **H** are in phase, mutually perpendicular, and perpendicular to the direction of propagation; the three other graphs are the projections of the wave on the XY, XZ and YZ planes.

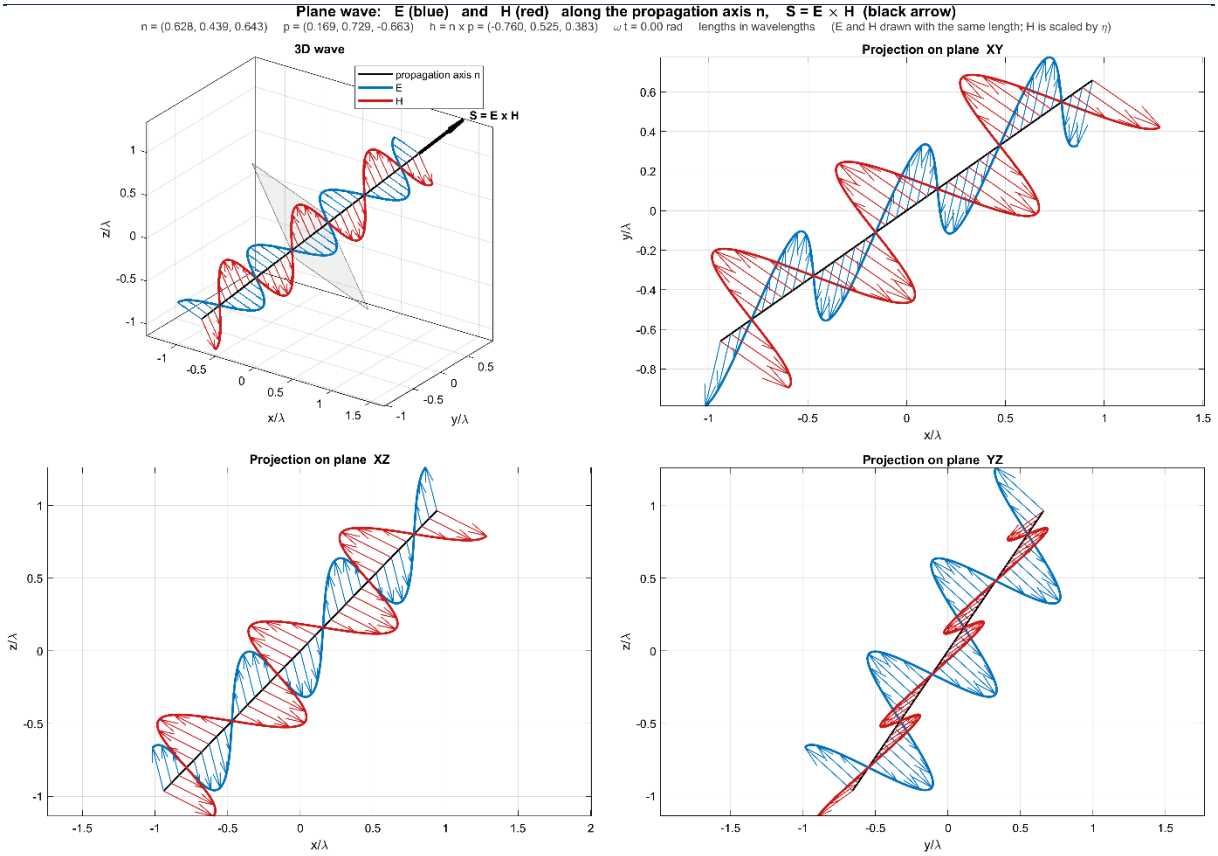


Figure 16. The plane wave in 3D: **E** (blue) and **H** (red) along the propagation axis **n**, one wavefront, the Poynting vector **S**, and the projections on the three coordinate planes.

Figure 17 isolates the plane perpendicular to the direction of propagation, spanned by **p** and **h**. The upper-left graph repeats the 3D wave with this transverse plane drawn as a dashed square. In the plane itself, **E** alone is a set of identical arrows along **p**, and **H** alone is a set of identical arrows along **h**; this uniformity is the defining feature of a plane wave, since the fields vary only along **n**. In the last graph the two fields are superimposed: they are at right angles, and their cross product **E** × **H** points out of the page, along +**n**. As time passes the arrows oscillate between $\pm E_0$ and $\pm H_0$ without changing direction, i.e. the wave is linearly polarised.

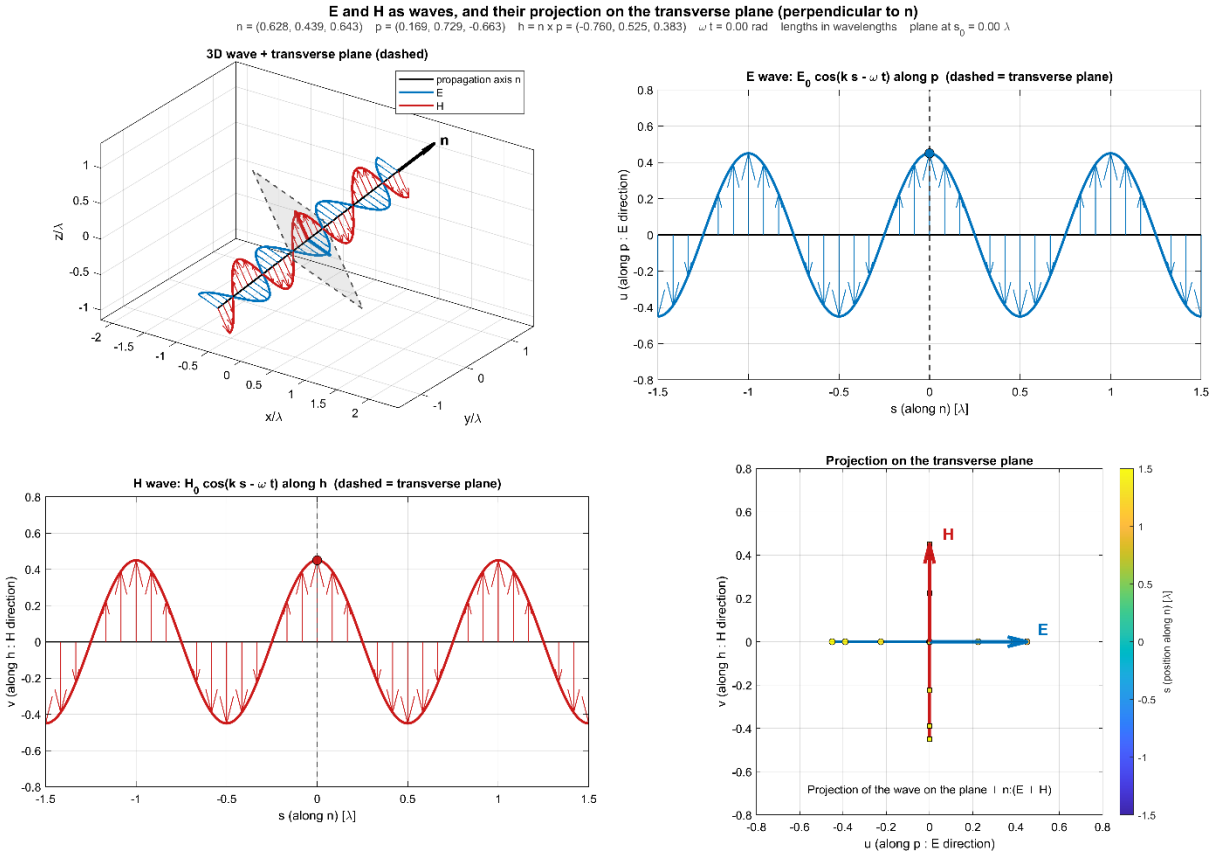


Figure 17. The wave and the transverse plane. Upper left: 3D wave with the plane perpendicular to **n**. Upper right: **E** only. Lower left: **H** only. Lower right: **E** and **H** together with **S** pointing toward the reader.

> **Key result.** For the uniform plane wave, $\nabla\cdot\mathbf{E} = \nabla\cdot\mathbf{H} = 0$; $\nabla\times\mathbf{E} = -\mu_0\partial\mathbf{H}/\partial t$ is parallel to **H** and $\nabla\times\mathbf{H} = \varepsilon_0\partial\mathbf{E}/\partial t$ is parallel to **E**, both in quadrature with the fields; the gradients of $|\mathbf{E}|$ and $|\mathbf{H}|$ are parallel to **n**; and **E**, **H** and **n** form a right-handed orthogonal triad with **S** = **E** × **H** along +**n**.

6. Conclusions

The three examples studied in this report show how divergence, curl and gradient characterise a field:

- The rotational field $\mathbf{F} = (-y, x, 0)$ has $\nabla\cdot\mathbf{F} = 0$ and $\nabla\times\mathbf{F} = 2\mathbf{a}_z$. Its rotation is visible only on the XY plane, where the curl component normal to the plane is non-zero, and Stokes' theorem is satisfied with a circulation of 2π on the unit circle.
- The saddle field $\mathbf{F} = (x, -y, 0)$ has $\nabla\cdot\mathbf{F} = 0$ and $\nabla\times\mathbf{F} = 0$. It derives from the harmonic potential $\varphi = (x^2 - y^2)/2$, and the circulation and the net flux vanish on every closed contour. Its magnitude equals that of the rotational field, which shows that the magnitude alone does not determine the divergence or the curl.
- For a uniform plane wave in free space propagating along an arbitrary direction **n**, both **E** and **H** are divergence-free, their curls are in quadrature with the fields and obey Faraday's and Ampère's laws, the gradients of their magnitudes point along **n**, and the triad (**E**, **H**, **n**) is right-handed and orthogonal.
- Visualising each field both in 3D and on orthogonal planes is a powerful tool, but it must be interpreted with care: a 2D plot hides the component normal to its plane, and only the curl component along that normal describes circulation in the plane.

The same scripts can be reused for other fields or other wave parameters (direction, polarisation, phase) by changing the corresponding lines at the top of each file.

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[1] W. H. Hayt Jr. and J. A. Buck, *Engineering Electromagnetics*, McGraw-Hill.

[2] D. K. Cheng, *Field and Wave Electromagnetics*, Pearson.

[3] C. A. Balanis, *Advanced Engineering Electromagnetics*, Wiley.

[4] D. J. Griffiths, *Introduction to Electrodynamics*, Cambridge University Press.

[5] The MathWorks, Inc., MATLAB documentation: *divergence*, *curl*, *quiver3*, *exportgraphics*.